Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/26/2024
PI Name:	Schuerger, Andrew Ph.D.		
Project Title:	Microgravity Can Down-Regulate Host Resistance and thus May Up-Regulate Plant Disease Development in Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	schuerg@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	321-261-3774
Organization Name:	University of Florida, Gainesville		
PI Address 1:	Department of Plant Pathology		
PI Address 2:	505 Odyssey Way, Exploration Park		
PI Web Page:			
City:	Merritt Island	**State:**	FL
Zip Code:	321-261-3774	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	01/01/2022	**End Date:**	12/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
Contact Email:	gioia.massa@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per PI (Ed., 11/26/24).		
Key Personnel Changes/Previous PI:	1) Ms. Kylee Soltez, Dept. of Plant Pathology (Univ. of FL) has joined the project as a lab assistant to Dr. Andrew C. Schuerger (PI). Jordan Callaham (M.Ag.), with UF's Horticultural Sciences Department, and Chad Vanden Bosch of Redwire, Inc., have also been added to the project. All have made significant contributions to the project and they will be on numerous papers in regards to the flight experiment. 2) Dr. Vicken Aknadibossian (post-doc), Dept. of Horticulture, (Univ. of FL) has left the project to begin a new post-doc position at another university.		
COI Name (Institution):	Ferl, Robert Ph.D. (University of Florida, Gainesville) Paul, Anna-Lisa Ph.D. (University of Florida, Gainesville) Reed, David M.S. (Redwire, Inc.)		
Grant/Contract No.:	80NSSC22K0209		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space-faring nations are utilizing small plant-growth payloads in microgravity (micro-g) to develop the knowledge and technology infrastructures to advance the development of food production systems on other planetary bodies. As the use of small plant-growth payloads in micro-g continues, plant disease outbreaks will increase over time, once the systems are integrated into the open-air microbiomes of spacecraft. This situation presents an opportunity to address directly Section 2.3.2.B of NASA Solicitation 2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research – the combined effects of various space-relevant stressors – in a manner that further enables exploration. A solid literature base exists that indicates that plant host resistance is down-regulated in micro-g and includes studies that describe decreased cell wall rigidity, cell wall thickness, cellulose and matrix polysaccharides, lignin, and altered host-resistance gene pathways in micro-g. An equally solid literature base indicates that microbial virulence may be up-regulated in microgravity and includes up-regulation of virulence in microbe/microbe, microbe/insect, and pathogen/plant interactions. However, no data exists on the interactions of a foliar phytopathogen on a plant host with concomitant host-resistance transcriptomics data. The alternative hypothesis (Ha) for the International Space Station (ISS)-flight experiment is: Microgravity Can Down-Regulate Host Resistance and thus May Up-Regulate Plant Disease Development in Space. Results will fill key knowledge gaps into how plant diseases and host resistance are affected by micro-g. Proposed here is a novel flight experiment that will study the development of a foliar plant pathogen (i.e., phytopathogen) on the well-studied, Arabidopsis thaliana (At) host. The phytopathogen – Golovinomyces cichoracearum (Gc), a powdery mildew fungus - on A. thaliana is a well-studied pathosystem. The Gc/At pathosystem is chosen here because (i) both Gc and At are sequenced and annotated, (ii) extensive literature is available on host-resistance in At, (iii) diverse At cultivars with differential expression of easily measured host resistance mechanisms are available, (iv) most stages of the Gc life-cycle are on external surfaces of leaves and can be easily observed, (v) the expected ease of sanitizing flight hardware, and (vi) maximized crew safety on the ISS because Gc has no established interaction with humans. We will use the Multi-Use Variable-Gravity Platform (MVP) facilities built by Redwire, Inc. (previously Techshot, Inc.) (2 units are in orbit on the ISS) to investigate the development of Gc on leaves of At. Each MVP has two independently controlled centrifuge rotors fitted with up to 4 Phytofuge plant-growth modules that will be rotated at 1g or left stationary in micro-g. Each Phytofuge unit has three separate petri dishes with light-emitting diode (LED) illumination and an internal camera. Seed of two cultivars of At will be (1) sown onto growth media in independent petri dishes, (2) held dormant for up to 30 days, and (3) once in orbit, one-half of the petri dishes will be inoculated with the powdery mildew phytopathogen Gc. The aerial mycelia, conidiophores, and spores of Gc will be allowed to develop for 10-12 days and then leaves harvested for two separate research pipelines. First, half of the healthy and half of the infected At plants will be fixed in 3% buffered glutaraldehyde and stored at 4C until processed on the ground for fluorescent staining, Scanning Electron Microscopy (SEM), and Transmission Electron Microscopy (TEM) studies into the process of host infection. Second, the remaining healthy and infected plants will be fixed in RNAlater, frozen at -80C, and later processed for transcriptomics of host-resistance genes. Results will inform future horticulturists, space engineers, and technologists of the risks of maintaining plant-host resistance in space when challenged by an airborne phytopathogen. The results will also assist in the design of future plant-growth modules for crewed missions to the Moon and Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plant disease development in space has profound impact on the future of human exploration of the Solar System. Currently the assumption is that plants grown in space-based bioregenerative life support systems (BLSS) modules will develop normal plant-resistance mechanisms to exposure of biological agents (e.g., bacteria, fungi, viruses). If disease resistance is "normal" in space-based BLSS modules, the use of crops for food, oxygen, and water recycling will be a viable option for crewed habitats on the Moon and Mars. In contrast, if plant diseases develop more quickly in space than on Earth, new and unique plant production protocols may have to be developed. The research outlined in this project seeks to identify if "plant resistance" against a fungal phytopathogen in microgravity progresses normally in the mustard plant, Arabidopsis thaliana. The fungal pathogen has the general name of "powdery mildew", but the species name is Golovinomyces cichoracearum. Powdery mildew phytopathogens have no proven disease risk to humans, and thus, there is no health risk to the astronauts on the International Space Station (ISS) during the flight experiment. The hypothesis being tested here is (Ha): Microgravity can down-regulate plant host resistance, and thus, may up-regulate plant disease development in space. The primary benefit to Earth-based agriculture will be to identify how disease resistance mechanisms operate under the unusual conditions of microgravity. Results may identify how to improve disease resistance in field crops on Earth.
	The three major accomplishments for 2024 were: 1) We successfully completed two Phytofuge Experiment Module (PEM) tests in late 2023 and Jan/Feb 2024. Results indicate that both plants and disease developed nominally within the Multi-Use Variable-g Platform (MVP) Phytofuge Experiment Modules (PEM) Petri dishes. 2) A spore inoculation device called a Conidia Application Device (CAD) unit (nicknamed the Conidiator) was developed and tested. Initial results are very positive, and we successfully injected visibly observable quantities of Gc conidia into Petri dishes designed for the Gc/At flight experiment. 3) An Experiment Requirements Document (ERD) for the MVP-Plant-02 International Space Station (ISS) flight experiment was developed and completed. A Science Verification Test (SVT) Readiness Review (SVT RR) is scheduled for 10-Dec-2024. Other Accomplishments for 2024 are: 1) A flight-rated Petri dish was designed and manufactured for the SVT. The Phytofuge Experiment Module (PEM) can accommodate up to three Petri dishes. The new Petri dishes have been modified to allow the injection of Gc conidia

	during the flight.
Task Progress:	2) Nine fluorescent (FL) stains were tested on Gc-infected At leaves. We have chosen three FL stains for the SVT, Experiment Verification Test (EVT) and ISS flight experiments. These are: (a) Coomassie Brilliant Blue (CBB; stains Gc mycelia), (b) Aniline Blue (AB; stains plant callose), and (c) 3,3'-dihxyloxacarbocyanine iodide (DCI; used for whole canopy staining of aerial mycelia). 3) Electron microscopy (EM) fixation and processing protocols are 90% completed. A final test of the scanning electron microscopy (SEM) and transmission electron microscopy (TEM) fixation and handling process will be completed during the SVT (now scheduled for Jan/Feb 2025). 4) A leaf canopy protocol was developed using the program, ImageJ, to precisely estimate the leaf areas of plants growing within the PEM Petri dishes. The ImageJ protocol will be used in the SVT, EVT, and ISS flight experiment. 5) While working on testing SEM fixation protocols for infected leaves, we conducted a small scale experiment to determine if simulated microgravity could promote the development of the fungus, G. cichoracearum on At. Results indicated that germ tube lengths (i.e., a measure of early fungal growth rates) were significantly longer for conidia germinated on squash and At leaves incubated on 2D or 3D clinostats compared to 1-g controls. 6) We have scheduled the SVT for mid-January to mid-February 2025.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Schuerger AC, Podbielski M, Reed D, Paul A-L, Ferl, RJ. "Development of the phytopathogen, Golovinomyces cichoracearum (a powdery mildew), on Arabidopsis thaliana in microgravity: Preliminary ground studies towards an ISS flight experiment." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstract #34. , Dec-2023
Abstracts for Journals and Proceedings	Soltez K, Vanden Bosch C, Reed D, Logan S, Schuerger AC. "Development of an inoculation device for fungal plant pathogen research in space." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstract #4. , Dec-2024
Abstracts for Journals and Proceedings	Schuerger AC, Soltez KS, Kelley KL, Cooper QJ. "Clinorotation enhances growth of the biotrophic fungus Golovinomyces cichoracearum (a powdery mildew) on squash leaves." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstract #67. , Dec-2024